

Introduction to Machine Learning

What is Machine Learning?

Machine learning is a branch of artificial intelligence that allows computers to learn patterns from data and make predictions without being explicitly programmed for every task.

The system learns from examples and improves its performance as it receives more information.

How Machine Learning Works

A machine learning system usually follows three steps:

1. Collect data from the real world.
2. Train a model using examples.
3. Use the model to make predictions on new data.

The quality of the result depends on the quality of the data and the training process.

Common Applications

Machine learning is used in many areas:

- Recommendation systems suggest products and content.
- Image recognition identifies objects in pictures.
- Speech recognition converts spoken words into text.